Program #7: Develop a program to print the same above pattern by using space character

```
I/P:        O/P:
H           H
A               A
R                   R
I                       I
```

- Here we must use the space \s escape sequence character.
- It inserts always one space. If we want to place multiple spaces, we must use \s multiple times.

```
class NamePatternWithSlashS{
    public static void main(String[] args){
        System.out.println("H");
        System.out.println("\s\sA");
        System.out.println("\s\s\s\sR");
        System.out.println("\s\s\s\s\s\sI");
    }
}
```

Program #8: Develop a program to print the name with single and double quote on console

```
I/P:            O/P:
Hari Krishna    'Hari Krishna'
Hari Krishna    "Hari Krishna"
```

- Here must we use the escape sequence characters \' , \"

```
class NameInQuotes{
    public static void main(String[] args){
        System.out.println("'Hari Krishna'");
        System.out.println("\"Hari Krishna\"");

        System.out.println(" ' ");
        System.out.println(' " ');

        //System.out.println("""); // CE
        //System.out.println(' ' '); // CE
```

```
        System.out.println("\"");
        System.out.println('\' ');

    }
}
```

1. We can place single ' in double "" directly as " ' "
2. We can place double " in single ' ' directly as ' " '
3. We cannot place double " in double "" directly as """, we must use \"
   """ CE , "\"" allowed

4. We cannot place single ' in single ' ' directly as ' ' ', we must use \'
   ''' CE , '\'' allowed

<u>Rule :</u> Inside single quote we can place only one character

'a' , '\'' , '1' , '#' allowed

'abc' , '10' , '@3' not allowed

<u>Note:</u> In Python, multiple characters are allowed in ' ' or in "" or in """ """

But in Java, multiple characters are allowed only in " " or in """ """

- """ is used for placing text block. It means for storing multiple lines of text as one string value we must use """

  For example:

```
        System.out.println(
            """
                H
                A
                R
                I
            """
        );
```

- Rewrite NamePrinterWithStars program with Text block single Sopl()

Program #9: Develop a program to print file path
D:\abc\xyz\enames.txt

```
class FilePath{
    public static void main(String[] args){
        //System.out.println("D:\abc\xyz\enames.txt"); // CE
        System.out.println("D:\\abc\\xyz\\enames.txt");
```

```
        System.out.println("D:\new\text\sname.txt"); // No CE
        System.out.println("D:\\new\\text\\sname.txt");
    }
}
```

Escape Sequence characters

1. A character preceded by \ is called esc seq character
2. The \ either adds or removes the special meaning to the character placed after \.
3. The \ adds the special meaning to the letter n , t and s
   The \ removes the special meaning to the letter " , ' and \
4. After \ we cannot place a letter or a digit or specific character randomly as we like. From one PL to another PL the list will be changed
5. The Java supports below 18 sec seq characters (9+8+1)

Letter based esc seq characters

1. \n - inserts one new line
2. \t - inserts one tab spaces [less than or equals to 8 spaces]
3. \r - inserts one carriage return
4. \s - inserts one space
5. \f - inserts one form feed
6. \b - inserts one back space
7. \' - inserts one single quote '
8. \" - inserts one double quote "
9. \\ - inserts one \

Octal digits esc seq characters

1. \0 - inserts one null character (it is not empty character)
2. \1
3. \2
4. \3
5. \4
6. \5
7. \6
8. \7

UNICODE esc seq character

1. \uXXXX

Rule: In Java, After \ we are allowed to place only the above 18 characters.

If we place any other character, compiler will throw error illegal escape character